Solutions To Select Problems

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Problem 1

Sakurai 5.3

The Unperturbed States

We begin by writing out our Hamiltonian before the perturbation, denoted by $\mathcal{H}_0$:

$$\mathcal{H}_0 = -\frac{\hbar^2}{2m}\nabla^2 + V_0(\vec{r}). \tag{1}$$

The potential prescribed, $V_0(\vec{r})$ is that of the 2 dimensional particle in a box, of length L:

$$V_0(x, y) = \begin{cases} 0, & \text{for } 0 \leq x \leq L,\, 0 \leq y \leq L \\ \infty, & \text{otherwise} \end{cases} \tag{2}$$

To derive the wave-functions $|n, m\rangle$, simply use a separation of variables on the Schrödinger equation, with the appropriate boundary conditions:

$$\begin{aligned} -\frac{\hbar^2}{2m}X''(x) &= \varepsilon_n\, X(x) \\ -\frac{\hbar^2}{2m}Y''(y) &= \varepsilon_n\, Y(y), \end{aligned} \tag{3}$$

such that $X(0) = Y(0) = X(L) = Y(L) = 0$. We identify the usual sin and cos solutions to this problem, and after plugging in the necessary boundary conditions and normalizing:

$$X_n(x) = \begin{cases} \sqrt{\frac{2}{L}}\sin\frac{n\,\pi\,x}{L}, & \text{for } 0 \leq x \leq L,\, 0 \leq y \leq L \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $Y_m(y)$ is an identical function that can be obtained by simply substituting $x \to y,\ n \to m$. We have arrived at a complete and orthonormal set of energy eigenstates that spans our Hilbert space:

$$|n, m\rangle = \frac{2}{L}\sin\left(\frac{n\,\pi\,x}{L}\right)\sin\left(\frac{m\,\pi\,y}{L}\right). \tag{5}$$

Since these are energy eigenstates, we also note that $\mathcal{H}_0\,|n, m\rangle = (n^2 + m^2)\frac{\hbar^2\pi^2}{2mL^2}$. As a result, the ground state is given by $|1, 1\rangle$ and the first excited state is doubly degenerate, with both $|1, 2\rangle$ and $|2, 1\rangle$ fitting the bill. These are the 0^{th} order eigenfunctions as well.

On To Pertubring

We now proceed to compute the desired perturbations that arise as a result of the perturbing potential,

$$V_1(x,y) = \begin{cases} \lambda xy, & \text{for } 0 \le x \le L,\ 0 \le y \le L \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Having identified that the ground state is non-degenerate, we may immediately produce the first order correction to the ground state energy:

$$\begin{aligned} \Delta^{(1)}_{1,1} &= \langle 1,1|\, V_1\, |1,1\rangle \\ &= \lambda \frac{4}{L^2} \int_0^L \int_0^L dx\, dy\, \sin\left(\frac{\pi\, x}{L}\right) \sin\left(\frac{\pi\, y}{L}\right) x\, y \\ &= \lambda \frac{L^2}{4} \end{aligned} \tag{7}$$

We now proceed to the first excited state. This is slightly more complicated as we have 2 degenerate energy eigenstates. Consider now, this 2 state subspace of the Hilbert space we work in. We can provide an explicit form for the V_1 operator in this subspace. Let us denote $|\alpha_1\rangle \equiv |2,1\rangle$ and $|\alpha_2\rangle \equiv |1,2\rangle$. Then, the explicit form of V_1 in the α-subspace is:

$$V_1^\alpha = \begin{pmatrix} \langle\alpha_1|\, V_1\, |\alpha_1\rangle & \langle\alpha_1|\, V_1\, |\alpha_2\rangle \\ \langle\alpha_2|\, V_1\, |\alpha_1\rangle & \langle\alpha_2|\, V_1\, |\alpha_2\rangle \end{pmatrix} \tag{8}$$

To produce a numerical form (in terms of L, λ and constants of nature), we must explicitly evaluate these integrals (to avoid the tedium, I have employed Mathematica). The numerical form is:

$$V_1^\alpha = \lambda \frac{L^2}{4} \begin{pmatrix} 1 & \frac{1024}{81\pi^4} \\ \frac{1024}{81\pi^4} & 1 \end{pmatrix}$$

We need to diagonalize V_1^α and produce eigenvectors in the α subspace. To do so, we first compute the eigenvalues of V_1^α in the usual way. Solving the characteristic equation gives us the eigenvalues $\epsilon_\pm = \frac{\lambda L^2}{324\pi^4}(81\pi^4 \pm 1024)$. These eigenvalues are the first order energy shifts for the first excited state(s). With this, we conclude the solution.

Remarks

It is important to be able to produce the linear combination of the states that would form the "degeneracy-lifting" states, i.e. eigenstates that diagonalize the V_1^α operator. To do so, simply find normalized eigenvectors of V_1^α in the form $(a,b)^T$, and the corresponding eigenstates will simply be $a\,|\alpha_1\rangle + b\,|\alpha_2\rangle$, since we are working in the space of $\{|\alpha_1\rangle, |\alpha_2\rangle\}$. This is equivalent to writing $V_1^\alpha = P D_{V_1^\alpha} P^{-1}$, and then doing a basis transformation

$$\begin{pmatrix} |\alpha_1\rangle \\ |\alpha_2\rangle \end{pmatrix} \to P \begin{pmatrix} |\alpha_1\rangle \\ |\alpha_2\rangle \end{pmatrix}.$$

In our case, the eigenvectors of V_1^α are simply $\frac{1}{\sqrt{2}}(1,1)^T$ and $\frac{1}{\sqrt{2}}(1,-1)^T$, so the combination of states is simply $\frac{1}{\sqrt{2}}(|\alpha_1\rangle \pm |\alpha_2\rangle)$. In this case,

$$\begin{aligned} P &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ D_{V_1^\alpha} &= \begin{pmatrix} \epsilon_+ & 0 \\ 0 & \epsilon_- \end{pmatrix} \end{aligned}$$

is an equivalent descriptor.

To interpret this, we simply note that the first linear combination rises in energy by ϵ_+ to first order, while the second rises in energy by ϵ_- to first order, breaking the degeneracy.

Problem 2

Sakurai 5.26

The Unnormalized Force

The perturbing force takes the form

$$F(t) = \frac{F_0\tau/\omega}{(t^2+\tau^2)}, \qquad -\infty < x < \infty \tag{9}$$

Now, since we require a potential, we rely on the classical force, $V_x(t) = -F(t)x$, from the usual $dV = -\vec{F}\cdot d\vec{r}$. The probability for being in the first excited state is what we desire, given that the system is in the ground state alone at $t = -\infty$. I will only be computing this to the first order, so the only possible transition is $0 \to 1$, which is of $\mathcal{O}(F_0)$. The next order of correction is given by the processes $0 \to 1 \to 2 \to 1$ and $0 \to 1 \to 0 \to 1$, corrections of $\mathcal{O}\left(F_0^3\right)$. We use the formula provided in the Quantum Mechanics notes[1] for the probability of a first order transition[2]:

$$c_m(t) = \delta_{n1} - \frac{i}{\hbar}\int^t dt' e^{i\omega_{m1}t'} V_{m1}(t') \tag{10}$$

This assumes that $c_m(t_0) = \delta_{m1}$. However, we will depart slightly from this notation and use a 0 index for the ground state, modifying eq.(10) by changing $1 \to 0$. All we require is the matrix element $\langle 1|\, V_x(t)\,|0\rangle$ to compute this. Since the states are in the Heisenberg picture, it suffices to compute $\langle 1|\, x\,|0\rangle$ for this. To compute this, we rely on the Harmonic oscillator algebra that diagonalizes the Hamiltonian, noting that $x = \sqrt{\frac{\hbar}{2m\omega}}\left(a^\dagger + a\right)$. Clearly, we have the result:

$$\langle 1|\, V_x(t)\,|0\rangle = -F(t)\sqrt{\frac{\hbar}{2m\omega}} \tag{11}$$

We know that the oscillator energy levels are all equally spaced, seprated by $\hbar\omega$. Using eq.(10), we have:

$$c_1(t) = \frac{iF_0\tau}{\hbar\omega}\sqrt{\frac{\hbar}{2m\omega}}\int_{-\infty}^{t} dt' \frac{e^{i\omega t'}}{t'^2+\tau^2} \tag{12}$$

We identify this as the Fourier transform as we let $t \to \infty$, which is the limit required for the computation of the total probability. Identifying this Fourier transform gives us the result[3]:

$$c_1(\infty) = \frac{iF_0\tau}{\hbar\omega}\sqrt{\frac{\hbar}{2m\omega}}\left(2\pi\frac{e^{-|\omega|\tau}}{2\tau}\right) \tag{13}$$

The probability is simply the squared amplitude of c_1.

$$P^{(1)}(0\to 1) = \frac{\pi^2 F_0^2}{2\hbar\omega^2|\omega|}e^{-2|\omega|\tau} \tag{14}$$

Challenge for the Experts[4]

If F becomes normalized, it becomes a Cauchy distribution[5] with width determined by τ. This implies that the perturbation frequency is related to the inverse of this scale. To observe this, look at some normalized plots below:

[1] Set 2, written by Onuttom Narayan.
[2] Equation 2.6, Page 3.
[3] Gradshteĭn and Ryzhik, Page 1089, Table 12.23, Index 14.
[4] Sakurai really knows how to help your self-confidence.
[5] Also known commonly in the literature as a Lorentzian.

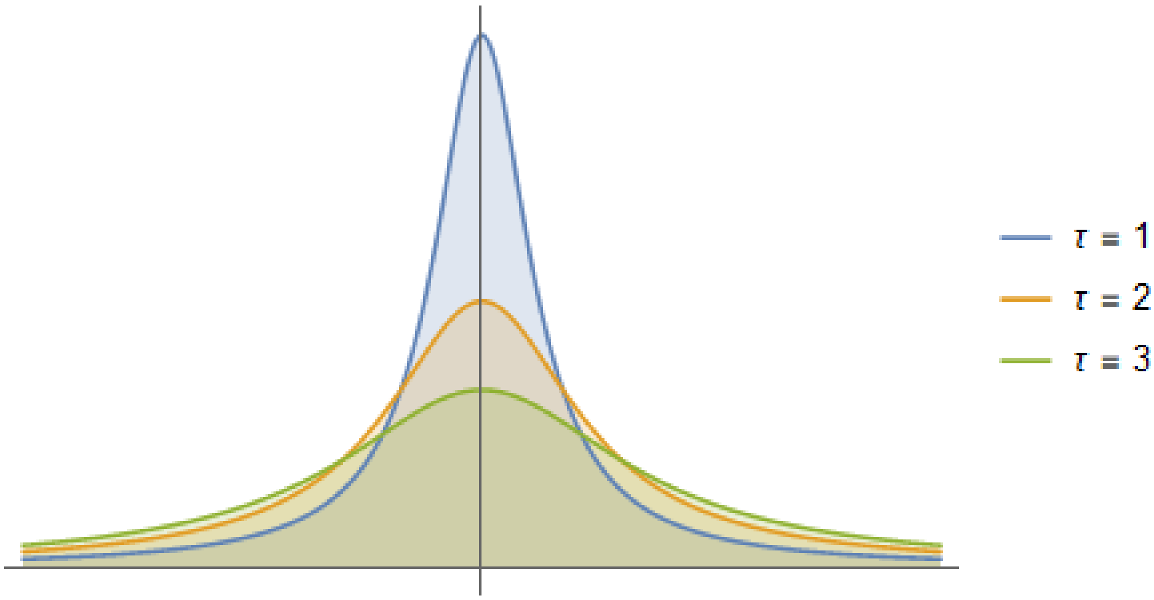

Figure 1: The distribution

Thus, if we took the limit of $\tau \gg \frac{1}{\omega}$, it is equivalent to turning on a perturbation very slowly, at a very low intensity (the peak depends on the width), and then undoing that. This is obviously a negligible perturbation in the limit, and would never, at any given point, be able to "perturb" the system out of the state (or more simply, if one were to hope for a well timed "2-photon-spectroscopy-like event", it would be suppressed by the sheer improbability of a large number of well ordered perturbations).

Problem 3

Sakurai 5.29

An Exact Solution

To solve this exactly, we need to work with states that are diagonal in the perturbing Hamiltonian, so that we may time evolve it. We have worked this system to the bone, and we know that the eigenbasis constitutes of linear combinations of the $|\pm\pm\rangle$ states. To use this fact, we may produce the rotated states, following the book's convention.

$$|1,1\rangle = |++\rangle\,, \quad |1,0\rangle = \frac{1}{\sqrt{2}}\left(|+-\rangle + |-+\rangle\right),$$
$$|1,-1\rangle = |--\rangle\,, \quad |0,0\rangle = \frac{1}{\sqrt{2}}\left(|+-\rangle - |-+\rangle\right)$$

We identify the initial condition as $|+-\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle + |0,0\rangle)$. The perturbing spin Hamiltonian is

$$H = \frac{4\Delta}{\hbar^2} S_1 \cdot S_2 = \frac{4\Delta}{\hbar^2}\left(S_1^z S_2^z + \frac{1}{2}S_1^+ S_2^- + \frac{1}{2}S_1^- S_2^+\right) \tag{15}$$

We identify the initial conditions as follows: $c_{10}(0) = c_{00}(0) = \frac{1}{\sqrt{2}}$. We now follow an explicit time evolution:

$$e^{-iHt/\hbar}|+-\rangle = \frac{1}{\sqrt{2}} e^{-iHt/\hbar}\left(|1,0\rangle + |0,0\rangle\right) = \frac{1}{\sqrt{2}}\left(e^{-i\Delta t/\hbar}|1,0\rangle + e^{3i\Delta t/\hbar}|0,0\rangle\right) \tag{16}$$

This comes from the spin calculations done with the expansion in eq.(15):

$$\frac{1}{\hbar^2} S_1 \cdot S_2 |+-\rangle = -\frac{1}{4}|+-\rangle + \frac{1}{2}|-+\rangle \tag{17}$$
$$\frac{1}{\hbar^2} S_1 \cdot S_2 |-+\rangle = -\frac{1}{4}|-+\rangle + \frac{1}{2}|+-\rangle \tag{18}$$

which allowed us to determine the results for $S_1 \cdot S_2 \frac{1}{\sqrt{2}}\left(|+-\rangle \pm |-+\rangle\right)$. Now, we note that we may express the results $c_{10}(t) = \frac{\exp(-i\Delta t/\hbar)}{\sqrt{2}}$ and $c_{00}(t) = \frac{\exp(3i\Delta t/\hbar)}{\sqrt{2}}$. This implies that $c_{11}(t) = c_{++}(t) = 0$ and $c_{1-1}(t) = c_{--}(t) = 0$, since $|1,1\rangle = |++\rangle$ and $|1,-1\rangle = |--\rangle$. Now we decompose the $|1,0\rangle$ and $|0,0\rangle$ back into the $|\pm\mp\rangle$ states, to get

$$c_{+-}(t) = \frac{1}{2}\left(e^{-i\Delta t/\hbar} + e^{3i\Delta t/\hbar}\right), \qquad c_{-+}(t) = \frac{1}{2}\left(e^{-i\Delta t/\hbar} - e^{3i\Delta t/\hbar}\right).$$

We may now square the coefficients to get the probabilities:

$$P(+-,t) = \cos^2\left(\frac{2\Delta t}{\hbar}\right), \qquad P(-+,t) = \sin^2\left(\frac{2\Delta t}{\hbar}\right), \qquad P(++,t) = P(--,t) = 0 \tag{19}$$

The Next Best Thing

While the Hamiltonian is not diagonal in the $\pm$ state basis, we note that the $|\pm\mp\rangle$ states form a closed subspace under the operation of the Hamiltonian. Thus, $\langle\pm\pm|\, H\, |\pm\mp\rangle$ vanishes identically, implying that, to first order, there are no allowed transitions to these states. We then only concern ourselves, to first order, with the transition $|+-\rangle \to |-+\rangle$. We use eq.(10) to compute this. The matrix element $\langle -+|\, H\, |+-\rangle$ can be computed by noting that eq.(17) and the orthonormality of the 2-spin states implies $\langle -+|\, S_1 \cdot S_2\, |+-\rangle = \hbar^2/2$. Thus, the matrix element is simply 2Δ. However, to compute the exponential term, we need to work in the diagonal eigenbasis of the Hamiltonian once more. This consideration can be simplified by noting that $|+-\rangle$ and $|-+\rangle$ are simply the positively ordered symmetric and anti-symmetric combinations of $|1,0\rangle$ and $|0,0\rangle$. Thus, the energy difference between these states is simply 0, given that they are eigenstates of the Hamiltonian. This can be explicitly verified by noting that eq.(17) and eq.(18) imply that $\langle +-| S_1 \cdot S_2 |+-\rangle = \langle -+| S_1 \cdot S_2 |-+\rangle$. Then we may state the transition amplitude:

$$c_{-+}(t) = -\frac{i}{\hbar}\int_0^t dt'\, 2\Delta \tag{20}$$

This is a trivial integral and provides the result $c_{-+}(t) = \frac{-2i\Delta}{\hbar}t$. Since we don't simply desire the amplitudes for transition, but wish to verify the validity of ”staying in place” with perturbation theory, we compute (with the obvious $\omega_{(+-)(+-)} = 0$):

$$c_{+-}(t) = 1 - \frac{i}{\hbar}\int_0^t dt'\,(-\Delta) \tag{21}$$

This allows us to present the probabilities:

$$P^{(1)}(+-,t) = 1 - \left(\frac{\Delta t}{\hbar}\right)^2, \qquad P^{(1)}(-+,t) = \left(\frac{2\Delta t}{\hbar}\right)^2, \qquad P^{(1)}(++,t) = P^{(1)}(--,t) = 0 \tag{22}$$

Note that $P^{(1)}(-+,t) \approx P(-+,t)$ for $t \ll \frac{\hbar}{2\Delta}$, but $P^{(1)}(+-,t)$ does not maintain its validity even for short times, implying the non-conservation of probability. So this would be a useful tool to compute the transition probability for short times.

Problem 4

Sakurai 7.5[6]

Solution

Since this is a symmetry of the theory, for any state $|\psi\rangle$, we require $e^{\mp i2\pi J_z/3}|\psi\rangle = \alpha|\psi\rangle$, i.e., $|\psi\rangle$ is an eigenvector of the rotation operator above. Since these are scalars, they are rotational invariants. Thus, $\alpha = 1$. Now, we may make this more explicit in the usual $|j,m\rangle$ basis: $e^{\mp i2\pi J_z/3}|j,m\rangle = |j,m\rangle$. Since $J_z|j,m\rangle = m|j,m\rangle$, we note that m has to take on a value such that $\mp 2\pi\frac{m}{3} = 2n\pi$ for some $n \in \mathbb{Z}$. Thus, $m = 3k$ for $k \in \mathbb{Z}$ is a requirement.

[6] 6.4 in the old version.

Problem 5

Sakurai 7.9[7]

Solution

We know that the states of the harmonic oscillator are given in a normalized position basis by $\langle x|n\rangle = \sqrt{\frac{2}{L}} \sin\frac{n\pi x}{L}$, with energies of $\frac{n^2\pi^2\hbar^2}{2mL^2}$. Further, we know fermionic wave functions must be anti-symmetric under interchange.

The Triplet Ground State

The triplet ground state is spin symmetric, which necessitates an anti-symmetric spatial wave function. Thus, we conclude that the ground state for the triplet will be given by $\langle x_1, x_2| \frac{1}{\sqrt{2}}(|1,2\rangle - |2,1\rangle)$, where $|i,j\rangle = |i\rangle \otimes |j\rangle$. More explicitly,

$$\langle x_1 x_2|ground\rangle_{triplet} = \frac{\sqrt{2}}{L}\left(\sin\frac{\pi x_1}{L}\sin\frac{2\pi x_2}{L} - \sin\frac{2\pi x_1}{L}\sin\frac{\pi x_2}{L}\right) \tag{23}$$

The energies can be obtained in the usual way, by operating with $-(\frac{\nabla_1^2}{2m} + \frac{\nabla_2^2}{2m})$, giving us a total energy of $(2^2+1^2)\frac{\pi^2\hbar^2}{2mL^2} = \frac{5\pi^2\hbar^2}{2mL^2}$.

The Singlet Ground State

We follow the same reasoning, but assert that the anti-symmetric nature of the singlet state under interchange imposes that the spatial wavefunction be symmetric under interchange, implying that a $|1,1\rangle$ state is perfectly acceptable ground state. We may then provide a position basis expression:

$$\langle x_1 x_2|ground\rangle_{singlet} = \frac{2}{L}\left(\sin\frac{\pi x_1}{L}\sin\frac{\pi x_2}{L}\right) \tag{24}$$

The energy of this state is simply $(1^2+1^2)\frac{\pi^2\hbar^2}{2mL^2} = \frac{\pi^2\hbar^2}{mL^2}$.

A Perturbation

Note that the perturbing potential is spatially symmetric and entirely degenerate (proportional to the identity), and that the triplet state is spatially anti-symmetric. Thus, we immediately note that this potential cannot perturb the triplet state into a different energy level. Essentially, since $x_1 = x_2$ is Pauli excluded, they never feel the potential, which requires exactly the condition $x_1 = x_2$.
Since the singlet is spatially symmetric, we may use the first order correction formula $\Delta_n^{(1)} = \langle n| V |n\rangle$. This is tantamount to performing the integral:

$$\Delta^{(1)}_{singlet} = -\lambda \int_0^L dx\, \frac{4}{L^2}\sin^4\frac{\pi x}{L} = -\lambda\frac{3}{2L} \tag{25}$$

[7] 6.7 in the old version.

Problem 6

A particle with spin-1 has a Hamiltonian that is independent of spatial variables:

$$H = AS_z^2 + B(S_x^2 - S_y^2) \tag{26}$$

where $A \gg B$. Treating the B term as a perturbation, calculate the energy levels to second order in B and compare with the exact result.

The Hamiltonian

We require representations of group[8] $Spin(3)$, or $SU(2)$. It can be shown that a map from $SO(3)$ to $SU(2)$ could be reparametrized such that for any element of $SO(3)$, there are two pre-images in $SU(2)$. Such a parametrization would imply that each fibre was of cardinality 2, which implies that $SU(2)$ can be parametrized as a 2-fold covering of $SO(3)$. We will choose our representation such that our third generator is proportional to $\mathrm{diag}(1, 0, -1)$, and such that we have states

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad |-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{27}$$

From this, we construct S_+ with the usual normalisation convention[9] adopted in accord with the Clebsch-Gordan tables $S_+ |j, m\rangle = \sqrt{(j-m)(j+m+1)}\, |j, m+1\rangle$ for $m < j$, and 0 if $m = j$. It is also easily noted that $S_- = S_+^\dagger$. This trivially generates the matrix representations:

$$S_+ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}, \qquad S_- = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix} \tag{28}$$

Finally, this allows us to construct S_x and S_y using the relations $2S_x = S_+ + S_-$ and $2iS_y = S_+ - S_-$:

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \qquad S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{29}$$

Now that we have explicit forms for all of the spin operators, we may explicitly construct our unperturbed Hamiltonian along with the perturbing potential:

$$H_0 = A \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad V = B \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{30}$$

Clearly, the unperturbed Hamiltonian is diagonal in the basis of choice, but the full Hamiltonian is not. If we were to diagonalize the full Hamiltonian exactly, we would get diagonal elements of $\{0, A \pm B\}$, which is found quite easily by solving the characteristic polynomial of the full Hamiltonian for its eigenvalues.

The Perturbative Treatment

Clearly, the unperturbed Hamiltonian is degenerate, with two eigenvalues of A and one eigenvalue of 0. It is also easy to note that the states corresponding to the degeneracy are $|\pm 1\rangle$, which implies that the projection onto the degenerate subspace is given by $P = |1\rangle\langle 1| + |-1\rangle\langle -1|$. Let us now rotate in the sub-space spanned by P. Note that the matrix for V shows an equal treatment of

[8]Since this is an informal discussion about simply connected Lie groups, it doesn't matter if I interchange the terms 'group' and 'algebra'.

[9]In addition to the convention $\hbar = 1$.

the interchange between $|1\rangle \rightarrow |-1\rangle$ and $|1\rangle \leftarrow |-1\rangle$. This implies that we seek symmetric and anti-symmetric combinations of $|1\rangle$ and $|-1\rangle$. This is produced by the rotation matrix:

$$R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{pmatrix} \tag{31}$$

We note that H_0 is unchanged by rotations in the degenerate subspace (as it is identical in energy), i.e. $RH_0R^T = H_0$. We compute V_R in this rotated basis using RVR^T, given that the states are rotated by $R\,|j\rangle$.

$$V_R = RVR^T = \begin{pmatrix} B & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -B \end{pmatrix} \tag{32}$$

In the language we see in Sakurai, we have found $|l^{(0)}\rangle$, and it turned out to work well for $|l\rangle$. Then, as per equation 5.2.11, which states that

$$\Delta_l^{(1)} = \langle l^{(0)}|\, V\, |l^{(0)}\rangle$$

Clearly, the first order shifts will be $\pm B$ for the degenerate states, and 0 for the non-degenerate. This matches the exact answer (by virtue of the fact that $|l^{(0)}\rangle = |l\rangle$). Thus, we expect the second order perturbations to vanish, and a glance at the formula 5.2.15,

$$\Delta_l^{(2)} = \sum_{k \notin D} \frac{|V_{kl}|^2}{E_D^{(0)} - E_k^{(0)}}$$

confirms this. To see this, note that $k \notin D$ implies that we sum over the states outside the degenerate manifold. This only involves $|0\rangle$, or the second row of V_R, which is entirely vanishing. For the rest, an explicit computation will prove the same, though the diagonalization of the subspace guaranteed this. Thus, the energy shifts to the second order move us to $A + B$, 0 and $A - B$ in the energy levels.

The method I have used is akin to diagonalizing the perturbative Hamiltonian in the degenerate subspace, which was necessitated by a "bad" choice of basis that didn't allow the degeneracy to be lifted at first order. My method used a diagonalization of the subspace by knowledge of the symmetric and anti-symmetric combinations that would diagonalize it.